Pruning Methods Evaluation

PMSM EWMA Baselines - 8 Pruning Methods, 3 Datasets

1. Methodology

Three baseline models were retrained by Simon on an expanded feature set: EWMA (exponentially-weighted moving average) features at six time spans (60/300/1200/3000/6000/12000 samples) plus four physics-derived features (i_squared, power_dq, torque_speed, i_cross), bringing input dimensionality from 8-11 up to 78-81. Train/val/test splits use a diversity-based session split: test sessions chosen at random, train sessions chosen greedily to maximise pairwise decorrelation, remaining sessions to validation - addressing the discovery that ~30 of the 69 drive sessions were highly correlated with each other under the original random split.

Dataset	Inputs	Architecture	Params	Baseline MSE
PMSM_no_temps_ewma	78	2x64	9281	3.905e-3
PMSM_no_temps_pm_ewma	79	5x128	76417	1.830e-3
PMSM_all_in_ewma	81	4x64	17793	8.475e-4

Each of the three baselines was pruned with 8 methods (OBD, OBS, magnitude, structured/neuron, activation-based, redundancy, low-rank, Taylor first-order), sweeping sparsity from 10% to 90% in 10% steps, fine-tuning after each pruning step, and stopping early if test MSE exceeded 3x the unpruned baseline. Fine-tuning used a random 50,000-row subsample of the training set (not the full ~150k-970k rows) for computational tractability; test evaluation always used the full, untouched test set.

Methods excluded from this run

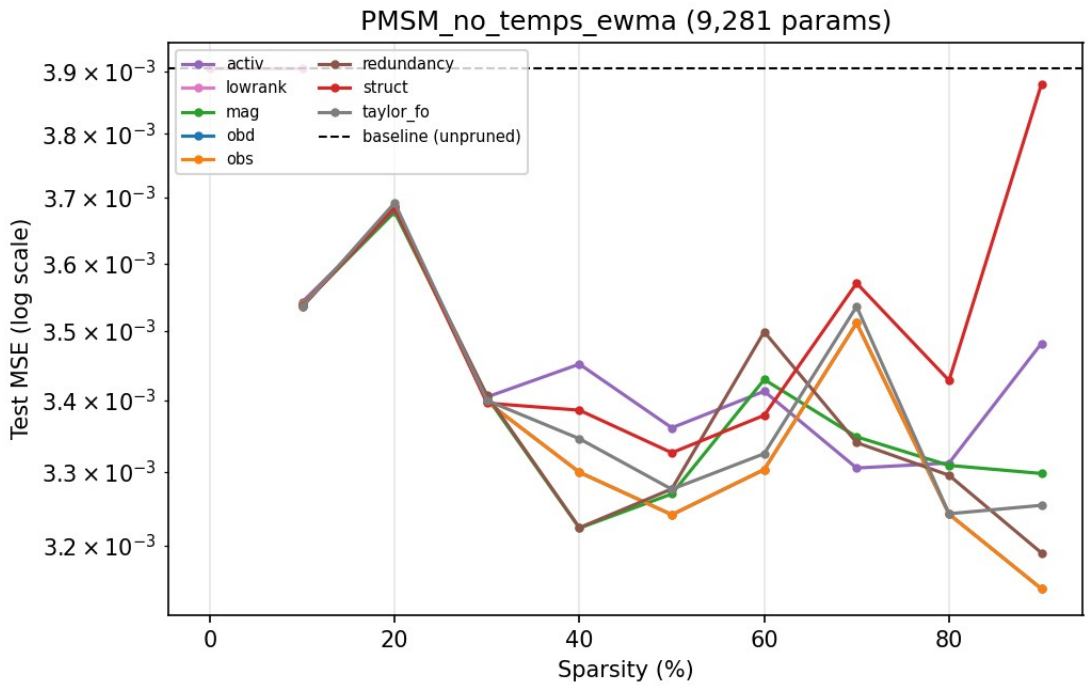
- SNIP and IMP+Rewind were not run on these baselines. Both require the model's state before training (SNIP prunes at initialization; IMP+Rewind prunes then rewinds surviving weights to their initial values before retraining). No training script in this repository saves that pre-training initialization - only final trained weights are saved - so there was no valid init to use. Running them with a freshly-seeded random init (unrelated to the model's actual starting point) was judged not meaningfully informative and was dropped by team decision.
- This is a single run per (method, dataset, sparsity) combination at a fixed seed (no repeated trials).

2. Results by Dataset

For each dataset, the table below lists every method's test MSE, MSE ratio vs. the unpruned baseline, remaining parameters, and FLOPs at every sparsity level tested (10% to 90%, in 10% steps). The chart shows the same data as MSE-vs-sparsity curves for all 8 methods.

PMSM_no_temps_ewma

Baseline (unpruned): test MSE = 3.905e-3, 9281 params, 9152 FLOPs.



Method	Sparsity	Test MSE	MSE Ratio vs Baseline	Params Left	FLOPs
obd	10.0%	3.538e-3	0.91x	8352	8227
obd	20.0%	3.685e-3	0.94x	7424	7306
obd	30.0%	3.398e-3	0.87x	6496	6386
obd	40.0%	3.300e-3	0.85x	5568	5465
obd	50.0%	3.242e-3	0.83x	4640	4556
obd	60.0%	3.303e-3	0.85x	3712	3643
obd	70.0%	3.512e-3	0.90x	2784	2724
obd	80.0%	3.242e-3	0.83x	1856	1809
obd	90.0%	3.143e-3	0.80x	928	913
obs	10.0%	3.538e-3	0.91x	8352	8227
obs	20.0%	3.685e-3	0.94x	7424	7306
obs	30.0%	3.398e-3	0.87x	6496	6386
obs	40.0%	3.300e-3	0.85x	5568	5465
obs	50.0%	3.242e-3	0.83x	4640	4556
obs	60.0%	3.303e-3	0.85x	3712	3643
obs	70.0%	3.512e-3	0.90x	2784	2724
obs	80.0%	3.242e-3	0.83x	1856	1809
obs	90.0%	3.143e-3	0.80x	928	913
mag	10.0%	3.538e-3	0.91x	8352	8227
mag	20.0%	3.678e-3	0.94x	7424	7317
mag	30.0%	3.404e-3	0.87x	6496	6399
mag	40.0%	3.224e-3	0.83x	5568	5480
mag	50.0%	3.270e-3	0.84x	4640	4565
mag	60.0%	3.430e-3	0.88x	3712	3648
mag	70.0%	3.349e-3	0.86x	2784	2724
mag	80.0%	3.309e-3	0.85x	1856	1802
mag	90.0%	3.298e-3	0.84x	928	881
redundancy	10.0%	3.537e-3	0.91x	8352	8226
redundancy	20.0%	3.682e-3	0.94x	7424	7311
redundancy	30.0%	3.407e-3	0.87x	6496	6396

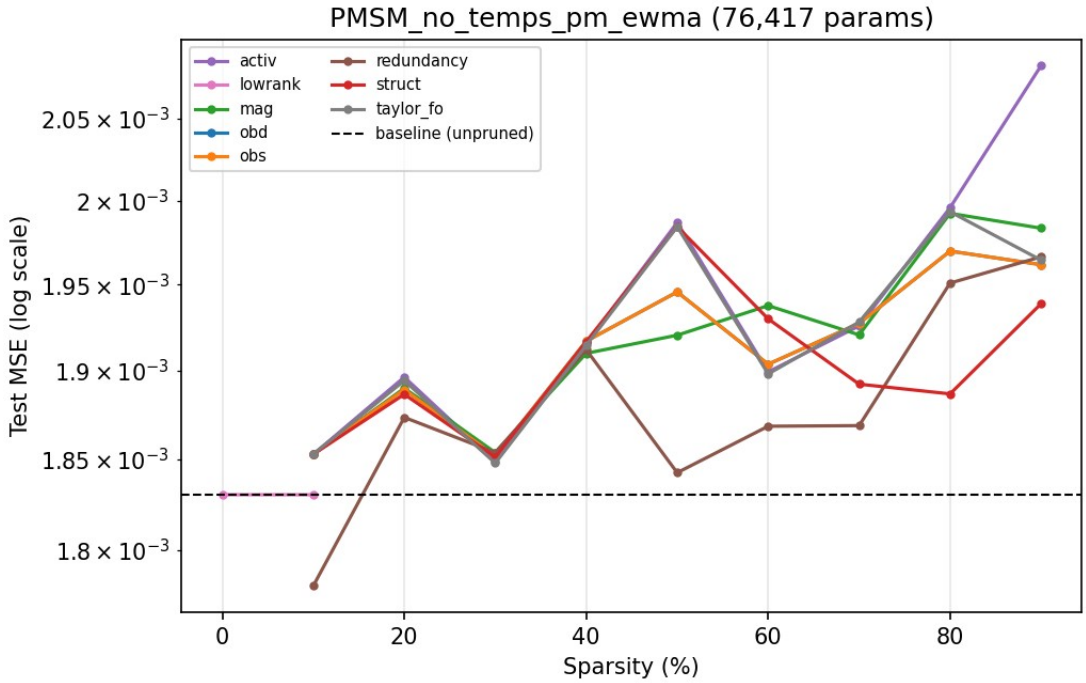
redundancy	40.0%	3.224e-3	0.83x	5568	5480
redundancy	50.0%	3.277e-3	0.84x	4640	4563
redundancy	60.0%	3.499e-3	0.90x	3712	3643
redundancy	70.0%	3.341e-3	0.86x	2784	2721
redundancy	80.0%	3.295e-3	0.84x	1856	1800
redundancy	90.0%	3.190e-3	0.82x	928	881
struct	10.0%	3.535e-3	0.91x	8337	8218
struct	20.0%	3.688e-3	0.94x	7378	7266
struct	30.0%	3.396e-3	0.87x	6419	6314
struct	40.0%	3.386e-3	0.87x	5460	5362
struct	50.0%	3.326e-3	0.85x	4603	4512
struct	60.0%	3.379e-3	0.87x	3613	3534
struct	70.0%	3.570e-3	0.91x	2768	2706
struct	80.0%	3.428e-3	0.88x	1845	1812
struct	90.0%	3.879e-3	0.99x	896	882
taylor_fo	10.0%	3.535e-3	0.91x	8352	8227
taylor_fo	20.0%	3.692e-3	0.95x	7424	7302
taylor_fo	30.0%	3.400e-3	0.87x	6496	6378
taylor_fo	40.0%	3.346e-3	0.86x	5568	5456
taylor_fo	50.0%	3.276e-3	0.84x	4640	4546
taylor_fo	60.0%	3.325e-3	0.85x	3712	3633
taylor_fo	70.0%	3.536e-3	0.91x	2784	2713
taylor_fo	80.0%	3.243e-3	0.83x	1856	1807
taylor_fo	90.0%	3.254e-3	0.83x	928	913
activ	10.0%	3.542e-3	0.91x	8291	8175
activ	20.0%	3.680e-3	0.94x	7395	7293
activ	30.0%	3.405e-3	0.87x	6435	6348
activ	40.0%	3.452e-3	0.88x	5539	5466
activ	50.0%	3.361e-3	0.86x	4622	4562
activ	60.0%	3.413e-3	0.87x	3650	3602
activ	70.0%	3.305e-3	0.85x	2759	2722
activ	80.0%	3.312e-3	0.85x	1787	1762
activ	90.0%	3.482e-3	0.89x	896	882
lowrank	0.0%	3.905e-3	1.00x	17474	9152
lowrank	10.1%	3.905e-3	1.00x	17474	9152

Best result for this dataset:

Best Method	Sparsity	Test MSE	MSE Ratio vs Baseline	Params Left
obd	90.0%	3.143e-3	0.80x	928

PMSM_no_temps_pm_ewma

Baseline (unpruned): test MSE = 1.830e-3, 76417 params, 75776 FLOPs.



Method	Sparsity	Test MSE	MSE Ratio vs Baseline	Params Left	FLOPs
obd	10.0%	1.853e-3	1.01x	68775	68161
obd	20.0%	1.889e-3	1.03x	61133	60576
obd	30.0%	1.852e-3	1.01x	53491	52950
obd	40.0%	1.917e-3	1.05x	45850	45325
obd	50.0%	1.946e-3	1.06x	38208	37867
obd	60.0%	1.904e-3	1.04x	30566	30353
obd	70.0%	1.927e-3	1.05x	22925	22761
obd	80.0%	1.970e-3	1.08x	15283	15138
obd	90.0%	1.962e-3	1.07x	7641	7566
obs	10.0%	1.853e-3	1.01x	68775	68161
obs	20.0%	1.889e-3	1.03x	61133	60576
obs	30.0%	1.852e-3	1.01x	53491	52950
obs	40.0%	1.917e-3	1.05x	45850	45325
obs	50.0%	1.946e-3	1.06x	38208	37867
obs	60.0%	1.904e-3	1.04x	30566	30353
obs	70.0%	1.927e-3	1.05x	22925	22761
obs	80.0%	1.970e-3	1.08x	15283	15138
obs	90.0%	1.962e-3	1.07x	7641	7566
mag	10.0%	1.853e-3	1.01x	68775	68144
mag	20.0%	1.890e-3	1.03x	61133	60718
mag	30.0%	1.854e-3	1.01x	53491	53155
mag	40.0%	1.910e-3	1.04x	45850	45610
mag	50.0%	1.920e-3	1.05x	38208	38060
mag	60.0%	1.938e-3	1.06x	30566	30450
mag	70.0%	1.921e-3	1.05x	22925	22834
mag	80.0%	1.992e-3	1.09x	15283	15202
mag	90.0%	1.983e-3	1.08x	7641	7575
redundancy	10.0%	1.781e-3	0.97x	68775	68144
redundancy	20.0%	1.873e-3	1.02x	61133	60591
redundancy	30.0%	1.854e-3	1.01x	53491	53095

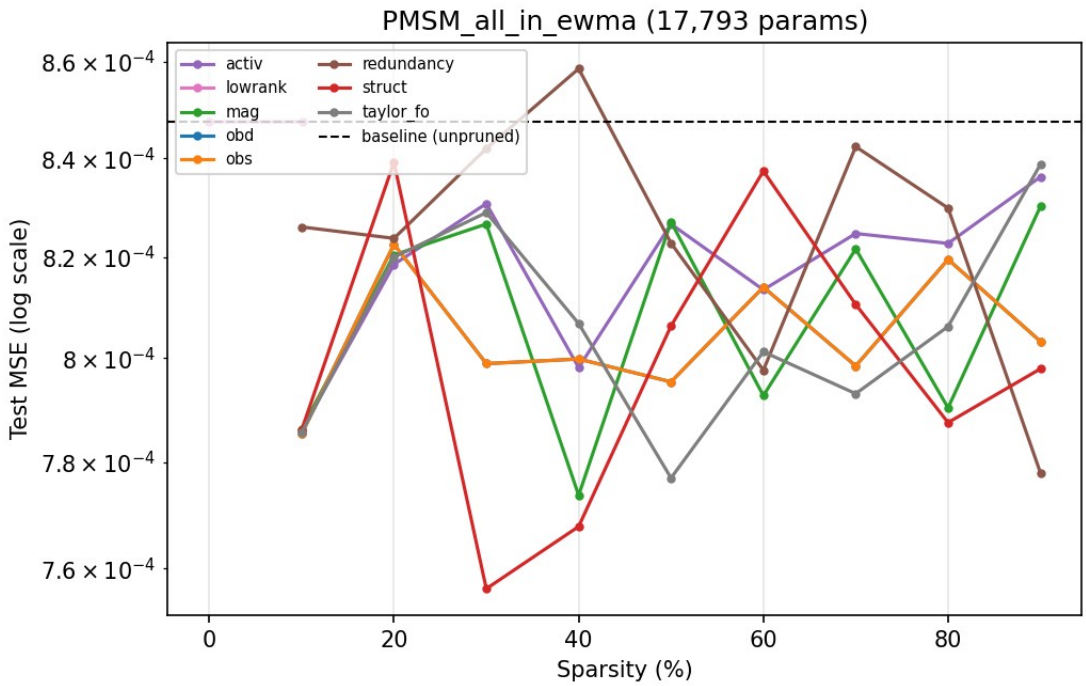
redundancy	40.0%	1.912e-3	1.04x	45850	45521
redundancy	50.0%	1.842e-3	1.01x	38208	37957
redundancy	60.0%	1.868e-3	1.02x	30566	30367
redundancy	70.0%	1.869e-3	1.02x	22925	22785
redundancy	80.0%	1.951e-3	1.07x	15283	15163
redundancy	90.0%	1.966e-3	1.07x	7641	7583
struct	10.0%	1.853e-3	1.01x	68694	68096
struct	20.0%	1.887e-3	1.03x	60998	60437
struct	30.0%	1.851e-3	1.01x	53302	52778
struct	40.0%	1.917e-3	1.05x	45801	45317
struct	50.0%	1.984e-3	1.08x	38106	37675
struct	60.0%	1.930e-3	1.05x	30500	30124
struct	70.0%	1.892e-3	1.03x	22917	22600
struct	80.0%	1.887e-3	1.03x	15270	15014
struct	90.0%	1.939e-3	1.06x	7611	7424
taylor_fo	10.0%	1.853e-3	1.01x	68775	68144
taylor_fo	20.0%	1.894e-3	1.04x	61133	60505
taylor_fo	30.0%	1.848e-3	1.01x	53491	52875
taylor_fo	40.0%	1.915e-3	1.05x	45850	45250
taylor_fo	50.0%	1.984e-3	1.08x	38208	37618
taylor_fo	60.0%	1.898e-3	1.04x	30566	30087
taylor_fo	70.0%	1.928e-3	1.05x	22925	22574
taylor_fo	80.0%	1.993e-3	1.09x	15283	15006
taylor_fo	90.0%	1.964e-3	1.07x	7641	7510
activ	10.0%	1.853e-3	1.01x	68694	68096
activ	20.0%	1.896e-3	1.04x	61053	60506
activ	30.0%	1.849e-3	1.01x	53321	52817
activ	40.0%	1.916e-3	1.05x	45816	45380
activ	50.0%	1.987e-3	1.09x	38063	37706
activ	60.0%	1.899e-3	1.04x	30552	30256
activ	70.0%	1.926e-3	1.05x	22914	22675
activ	80.0%	1.996e-3	1.09x	15276	15094
activ	90.0%	2.083e-3	1.14x	7578	7468
lowrank	0.0%	1.830e-3	1.00x	148195	75776
lowrank	10.0%	1.830e-3	1.00x	148195	75776

Best result for this dataset:

Best Method	Sparsity	Test MSE	MSE Ratio vs Baseline	Params Left
redundancy	10.0%	1.781e-3	0.97x	68775

PMSM_all_in_ewma

Baseline (unpruned): test MSE = 8.475e-4, 17793 params, 17536 FLOPs.



Method	Sparsity	Test MSE	MSE Ratio vs Baseline	Params Left	FLOPs
obd	10.0%	7.853e-4	0.93x	16013	15760
obd	20.0%	8.226e-4	0.97x	14234	13988
obd	30.0%	7.990e-4	0.94x	12455	12224
obd	40.0%	7.999e-4	0.94x	10675	10469
obd	50.0%	7.954e-4	0.94x	8896	8711
obd	60.0%	8.140e-4	0.96x	7117	6945
obd	70.0%	7.985e-4	0.94x	5337	5208
obd	80.0%	8.195e-4	0.97x	3558	3452
obd	90.0%	8.033e-4	0.95x	1779	1727
obs	10.0%	7.853e-4	0.93x	16013	15760
obs	20.0%	8.226e-4	0.97x	14234	13988
obs	30.0%	7.990e-4	0.94x	12455	12224
obs	40.0%	7.999e-4	0.94x	10675	10469
obs	50.0%	7.954e-4	0.94x	8896	8711
obs	60.0%	8.140e-4	0.96x	7117	6945
obs	70.0%	7.985e-4	0.94x	5337	5208
obs	80.0%	8.195e-4	0.97x	3558	3452
obs	90.0%	8.033e-4	0.95x	1779	1727
mag	10.0%	7.863e-4	0.93x	16013	15766
mag	20.0%	8.204e-4	0.97x	14234	14045
mag	30.0%	8.266e-4	0.98x	12455	12284
mag	40.0%	7.737e-4	0.91x	10675	10540
mag	50.0%	8.271e-4	0.98x	8896	8794
mag	60.0%	7.927e-4	0.94x	7117	7030
mag	70.0%	8.216e-4	0.97x	5337	5257
mag	80.0%	7.904e-4	0.93x	3558	3490
mag	90.0%	8.302e-4	0.98x	1779	1729
redundancy	10.0%	8.261e-4	0.97x	16013	15766
redundancy	20.0%	8.238e-4	0.97x	14234	14013
redundancy	30.0%	8.420e-4	0.99x	12455	12279

redundancy	40.0%	8.586e-4	1.01x	10675	10522
redundancy	50.0%	8.228e-4	0.97x	8896	8776
redundancy	60.0%	7.977e-4	0.94x	7117	7013
redundancy	70.0%	8.424e-4	0.99x	5337	5249
redundancy	80.0%	8.299e-4	0.98x	3558	3486
redundancy	90.0%	7.779e-4	0.92x	1779	1722
struct	10.0%	7.861e-4	0.93x	16000	15757
struct	20.0%	8.392e-4	0.99x	14156	13926
struct	30.0%	7.562e-4	0.89x	12336	12119
struct	40.0%	7.678e-4	0.91x	10620	10420
struct	50.0%	8.063e-4	0.95x	8832	8647
struct	60.0%	8.374e-4	0.99x	7075	6912
struct	70.0%	8.106e-4	0.96x	5286	5146
struct	80.0%	7.875e-4	0.93x	3514	3398
struct	90.0%	7.979e-4	0.94x	1779	1740
taylor_fo	10.0%	7.856e-4	0.93x	16013	15758
taylor_fo	20.0%	8.200e-4	0.97x	14234	13980
taylor_fo	30.0%	8.290e-4	0.98x	12455	12204
taylor_fo	40.0%	8.068e-4	0.95x	10675	10433
taylor_fo	50.0%	7.770e-4	0.92x	8896	8672
taylor_fo	60.0%	8.013e-4	0.95x	7117	6908
taylor_fo	70.0%	7.931e-4	0.94x	5337	5165
taylor_fo	80.0%	8.062e-4	0.95x	3558	3431
taylor_fo	90.0%	8.388e-4	0.99x	1779	1728
activ	10.0%	7.860e-4	0.93x	15942	15707
activ	20.0%	8.185e-4	0.97x	14184	13976
activ	30.0%	8.308e-4	0.98x	12390	12206
activ	40.0%	7.982e-4	0.94x	10611	10450
activ	50.0%	8.266e-4	0.98x	8878	8740
activ	60.0%	8.135e-4	0.96x	7103	6990
activ	70.0%	8.247e-4	0.97x	5337	5247
activ	80.0%	8.227e-4	0.97x	3537	3467
activ	90.0%	8.362e-4	0.99x	1773	1735
lowrank	0.0%	8.475e-4	1.00x	34178	17536
lowrank	10.1%	8.475e-4	1.00x	34178	17536

Best result for this dataset:

Best Method	Sparsity	Test MSE	MSE Ratio vs Baseline	Params Left
struct	30.0%	7.562e-4	0.89x	12336

3. Baseline Comparison - Which Dataset Performs Best

Comparing the three baselines directly: PMSM_all_in_ewma has the lowest absolute error both before and after pruning (it includes temperature sensor inputs the other two configs omit), while PMSM_no_temps_ewma - the smallest model - benefits the most from pruning in relative terms (best achieved ratio 0.80x, i.e. a 20% MSE reduction from its unpruned baseline).

Dataset	Params	Baseline MSE	Best Achieved MSE	Best Method	Best Ratio
PMSM_all_in_ewma	17793	8.475e-4	7.562e-4	struct @ 30.0%	0.89x
PMSM_no_temps_pm_ewma	76417	1.830e-3	1.781e-3	redundancy @ 10.0%	0.97x
PMSM_no_temps_ewma	9281	3.905e-3	3.143e-3	obd @ 90.0%	0.80x

4. Cross-Dataset Method Comparison

Each cell below shows that method's best operating point on that dataset (highest sparsity that keeps MSE within 1.5x baseline): sparsity % on top, MSE ratio and FLOPs ratio (both vs. the unpruned baseline) below.

Method	no_temps	no_temps_pm	all_in
OBD	**90%** 0.80x 0.10F	**90%** 1.07x 0.10F	**90%** 0.95x 0.10F
OBS	**90%** 0.80x 0.10F	**90%** 1.07x 0.10F	**90%** 0.95x 0.10F
Magnitude	**90%** 0.84x 0.10F	**90%** 1.08x 0.10F	**90%** 0.98x 0.10F
Taylor FO	**90%** 0.83x 0.10F	**90%** 1.07x 0.10F	**90%** 0.99x 0.10F
Structured	**90%** 0.99x 0.10F	**90%** 1.06x 0.10F	**90%** 0.94x 0.10F
Activation	**90%** 0.89x 0.10F	**90%** 1.14x 0.10F	**90%** 0.99x 0.10F
Redundancy	**90%** 0.82x 0.10F	**90%** 1.07x 0.10F	**90%** 0.92x 0.10F
Low-Rank	**10%** 1.00x 1.00F	**10%** 1.00x 1.00F	**10%** 1.00x 1.00F

>=70% sparsity reached 40-69% sparsity reached <40% sparsity / not run best method for that dataset

Since MSE scales differ across datasets, methods are also compared using rank (1 = lowest MSE at that sparsity level/dataset, averaged over all sparsity levels x datasets) and AUC of the MSE-ratio-vs-sparsity curve (lower = degrades more gracefully across the full sparsity sweep, not just at one point).

Average rank (lower is better)

Rank	Method	Average Rank (1 = best)	# Comparisons
1	obd	3.67	27
2	obs	3.67	27
3	redundancy	3.93	27
4	mag	3.96	27
5	struct	3.98	27
6	taylor_fo	4.04	27
7	activ	4.98	27
8	lowrank	6.00	3

Note: low-rank pruning has far fewer comparable points (n=3) since its rank-based sparsity ceiling is reached almost immediately on these small hidden-layer widths (64-128 neurons) - its rank is not directly comparable to the other methods' full 9-point sweep.

Average AUC of MSE-ratio-vs-sparsity

Method	Avg AUC (MSE ratio vs sparsity)	Interpretation
obd	0.954	Degrades gracefully
obs	0.954	Degrades gracefully
mag	0.956	Degrades gracefully
redundancy	0.957	Degrades gracefully
taylor_fo	0.957	Degrades gracefully
struct	0.959	Degrades gracefully

activ	0.966	Moderate
lowrank	1.000	Little/no headroom

5. Key Findings

- OBD and OBS tie for the best average rank (3.67) and lowest AUC (0.954) across all three datasets - both use curvature/Hessian-based saliency, and both benefit from a subsampled Fisher-diagonal approximation implemented to make them tractable on parameter counts above their exact-method threshold (5,000 for OBD, 3,000 for OBS).
- On PMSM_all_in_ewma and PMSM_no_temps_ewma, several methods (OBD, OBS, redundancy, mag) actually reduce test MSE below the unpruned baseline at 90% sparsity (ratios of 0.80x-0.95x), suggesting mild overparameterization in these small networks - pruning is acting as an implicit regularizer, not purely a compression trade-off.
- PMSM_no_temps_pm_ewma (the largest model, 76,417 params) is the one dataset where all methods land above 1.0x ratio at 90% sparsity (1.07x-1.14x) - the larger architecture appears to need more of its capacity, so the same aggressive 90% sparsity costs more accuracy here than on the two smaller nets.
- Low-rank pruning plateaus almost immediately (~10% sparsity) on all three datasets. With hidden layers only 64-128 neurons wide, there is very little singular-value redundancy to exploit - this is a structural limitation of the architecture size, not a flaw in the method.
- Activation-based and structured (whole-neuron) pruning consistently rank last among the weight-capable methods, though they still reach 90% sparsity everywhere - they trade more accuracy for coarser, hardware-friendlier sparsity patterns (removing whole neurons rather than individual weights).